Ingest Payload Contract v1.2

Version	1.2 (supersedes 1.1)
Date	August 2026
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Status	For implementation. Reconciled against the deployed endpoint, 2026-08-24

The wire format for delivering one certification extraction into Cartograph.

This version was verified against the Cartograph repository: model validations, strong-parameter lists, and the migration history. It corrects several v1.0 assumptions that would have failed at runtime. Those corrections are listed in Section 14.

Delivered to `POST /webhooks/runpod_ocr_results`. Endpoint and authentication are defined in *Integration Endpoints and Authentication*. A fully populated example accompanies this document as *Sample Ingest Payload*.

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1. Three decisions this contract makes

The payload speaks Cartograph's vocabulary, not the engine's. Translation happens engine-side before sending. Cartograph validates `income_type` and `asset_type` against fixed lists, so the receiving service should reject an unknown value rather than guess. This keeps the Rails service thin and places the mapping on the side that holds the domain knowledge.

Members are addressed by client-assigned refs, never by name. Each member carries a `ref` invented by the engine (`"m01"`); income, asset, and expense records reference `member_ref`. The importer builds a ref → ID map as it creates members. Name matching across free-text strings is a known source of silent mis-attribution; addressing members by ref removes it.

Rows Cartograph already holds are addressed by their own ids. The request's `existing_records_manifest` lists the case's existing members, income records and asset records with their ids. A payload row that corresponds to one carries that id back (`existing_member_id` and `person_id` on a member, `existing_record_id` on an income or asset record) and the importer updates the row in place, filling blanks only; a row without an id is created. A missed link costs a duplicate for staff to merge, a wrong one writes one person's figures onto another's record, so a link needs identity evidence: the date of birth with a recognisable name, a close name alone when a date is missing, or a near-identical uncontested name when the dates disagree (reported). Record links need the member link plus the same type (Cartograph's aliases applied) and, among several rows of one type, the closest source name.

Data records are full-replace; findings are not. Extraction-owned child records are cleared and rewritten on re-import. Findings are matched on their stable `finding_key` and updated in place, so a re-audit does not duplicate them and reviewer resolutions survive. See Section 11.

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2. Envelope

```
{
  "schema_version": "1.2",
  "extraction_id":  "ext_01JQ4X...",     // echoed back in the result callback
  "engine_version": "idp-2.5.0",
  "extracted_at":   "2026-08-17T14:22:09Z",
```

```
  "case_ref":       "CAS600142",          // REQUIRED - Cartograph Job#ref_number
  "target": {
    "job_id":       4821,                 // optional, if known
    "community_id": 67,                   // optional cross-check; 409 on mismatch
    "unit_number":  "2-114"
  },

  "cert_review":        { ... },          // Section 3
  "household_members":  [ ... ],          // Section 4
  "income_records":     [ ... ],          // Section 5
  "asset_records":      [ ... ],          // Section 6
  "expense_records":    [ ... ],          // Section 7
  "findings":           [ ... ]           // Section 8 - carries item_code
}
```

`case_ref` is the only required routing field: Cartograph's controller resolves it with `Job.find_by(ref_number: case_ref)`. An unresolvable `case_ref` is recorded on the scan and reported back through the result callback rather than rejected outright. Cartograph derives the `cert_review` from the job; there is no create route, and one is auto-created on first access.

Deliberately excluded: per-page OCR provenance, page classification, and document grouping. These are engine diagnostics with no operational use in Cartograph; they stay engine-side and are linked by `extraction_id`. Income calculations are also excluded; see Section 12.

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3. cert_review

Scalar fields only. Omit any key with no value; do not send `null`, which would blank a staff-entered value.

```
{
  "cert_type":         "annual",
  "effective_date":    "2026-07-01",
  "move_in_date":      "2019-03-15",
  "unit_number":       "2-114",
  "hh_size":           2,
  "annual_income":     "46377.18",
  "annual_assets":     "3362.55",
  "head_of_household_name": "Marcus Halvorsen",
  "tenant_rent":       "865.00",
  "contract_rent":     "1180.00",
  "gross_rent":        "1255.00",
  "gross_rent_basis":  "contract_plus_allowance",
  "hud_gross_rent":    "1255.00",
  "tenant_gross_rent": "940.00",
  "utility_allowance": "75.00",
  "max_program_rent":  "1024.00",
  "rent_field_sources": {
    "tenant_rent":       { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
    "utility_allowance": { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
    "contract_rent":     { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
    "gross_rent":        { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
    "hud_gross_rent":    { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
    "tenant_gross_rent": { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
                           "derived_from": ["tenant_rent", "utility_allowance"] },
    "max_program_rent":  { "form_type": "HUD-50059", "source_job_document_id": 2621, "effective_date": "2026-07-01", "
  }
}
```

Per `rent_extraction_contract` v2 on the case request, the two gross rents are also sent under their own names, each only when the form's figures settle it: `hud_gross_rent` is contract rent + utility allowance (a 50059's box 31); `tenant_gross_rent` is the tenant's rent + utility allowance with no subsidy in it (a TIC's printed gross rent; on a 50059 the total tenant payment, derived from the two printed figures and marked `derived_from`). `rent_field_sources` names, for every rent figure sent, the form type (`HUD-50059`, `TIC`,

`RD-3560-8`, `HUD-LEASE`), the request document (`source_job_document_id`, from `documents[].job_document_id`) and the packet pages it was read from, and the certification's effective date. The engine never puts a HUD gross rent into `max_program_rent`: that field carries only a printed rent limit.

`gross_rent` is the figure the certification prints under that name, and the two programs define it differently: HUD (50059, model lease, RD 3560-8) prints contract rent + utility allowance, a tax-credit TIC prints tenant rent + utility allowance. `contract_rent` is sent where the form prints one (50059 field 29, the lease's "Contract Rent", 3560-8 note rate rent) and omitted where it does not. `gross_rent_basis` says which definition the form's own arithmetic settles — `contract_plus_allowance` or `tenant_plus_allowance` — and is omitted when the figures settle neither, so a consumer that keeps the two under separate fields can route the figure without guessing from the form type.

`cert_type` accepts `initial`, `annual`, `interim` only. The engine's four certification types map as `MI` → `initial`, `AR` → `annual`, `IR` → `interim`; **AR-SC has no target value** and is an open decision (Section 15).

`result` and `signed_off_at` are staff state and are never written by the importer.

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4. household_members

`ref` is required and must be unique within the payload. `first_name` and `last_name` are validated for presence.

```
{
  "ref":               "m01",
  "existing_member_id": 4629,              // manifest member id, or null (Section 1)
  "person_id":          1981,              // echoed from the manifest, or null
  "aka_names":          [],                // the packet's spelling when new to Cartograph; a suggestion
  "first_name":        "Marcus",
  "last_name":         "Halvorsen",
  "date_of_birth":     "1974-11-02",     // ISO; parsed engine-side
  "relationship":      "Head of Household",
  "is_hoh":            true,
  "is_disabled":       false,
  "full_time_student": false,
  "ssn_last4":         "3168",           // last four only, see Section 13
  "email":             "m.halvorsen@example.com",
  "cell_phone":        "555-0142",
  "sort_order":        1
}
```

The engine does not extract relationship-to-head. It sends `"Head of Household"` where the head flag is set and omits the key otherwise, leaving the field unset rather than inferring a value.

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5. income_records

One object per **(member, source)** pair. The engine's flat per-paystub output is grouped into these parents before sending; paystubs, VOIs, and zero-income affidavits nest inside.

```
{
  "member_ref":            "m01",
  "existing_record_id":    4329,                 // manifest income record id, or null
  "income_type":           "non_federal_wages",   // validated, Section 10.1
  "source_name":           "Northgate Logistics LLC",
  "verification_status":   "received",
  "frequency_of_pay":      "biweekly",
  "date_received":         "2026-06-18",
  "employment_start_date": "2019-04-01",
  "employer_address":      "1400 Depot Rd, Salem, OR 97301",
  "self_declared_amount":  "45900.00",           // the household's own figure
  "source_of_declaration": "Application",
  "certification_form":    "HUD-50059",          // what the certification itself states
```

```
  "certification_amount":  "45900.00",            // for this source, annual; or null

  "paystubs": [
    { "pay_date": "2026-06-05", "gross_pay": "1742.10",
      "ytd_amount": "21071.75", "pay_frequency": "biweekly" }
  ],
  "vois": [
    { "voi_type": "Employer Verification", "date_received": "2026-06-18",
      "rate_of_pay": "21.78", "hours_per_pay_period": "80.00",
      "frequency_of_pay": "biweekly", "ytd_amount": "18406.00",
      "ytd_start_date": "2026-01-01", "ytd_end_date": "2026-06-05" }
  ],
  "eivs": [],
  "zero_income": null
}
```

An EIV income report is its own child row. A record verified by HUD's EIV income report carries one `eivs[]` row per benefit section of the report, attached to that benefit's record (the Social Security section on the Social Security record, the SSI section on the SSI record), never a `vois` row for the same pages:

```
"eivs": [
  { "benefit_type": "ssi", "source_name": "Social Security Administration",
    "report_date": "2026-02-05", "amount": "994.00", "frequency": "monthly",
    "annual_amount": null, "payment_status": "C01 - Current Pay",
    "notes": null, "pages": [12, 13] }
]
```

`benefit_type` is one of `social_security`, `ssi`, `dual_entitlement`, `employment_wages`, `unemployment_benefits`, `other`. `report_date` is the date of the report itself, not the benefit's start: the report can list a benefit since discontinued, and the date is what places the amount in the certification year. `amount` is the periodic figure in force with its `frequency`; `annual_amount` only when the report states one. `payment_status` is the section's code as printed. A section reporting no data, or a denied or terminated claim with no current amount, is not income and produces no record.

The certification's figure is not a self-declaration. `self_declared_amount` is what the household wrote on its own application, questionnaire or affidavit, with `source_of_declaration` naming the form. What the certification form itself prints for the source goes out on `certification_amount`, annual, under `certification_form` (`HUD-50059` or `TIC`); Cartograph compares it with its own calculation and raises an issue when they differ. A figure read off the certification is never sent as self-declared. A source the form lists and nothing in the packet verifies is its own record (`verification_status: declared_only`) carrying the certified amount, so there is a row to attach it to. A certification Cartograph does not recognise by name (RD 3560-8) carries no certified figures.

Zero income is a child, not a type. `zero_income` is not a valid `income_type` and fails validation. Set the parent to `other` and attach a `zero_income` object with `affidavit_date`, `signed`, `notes`.

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6. asset_records

```
{
  "member_ref":                "m01",
  "existing_record_id":        3519,           // manifest asset record id, or null
  "asset_type":                "checking",     // validated, Section 10.2
  "institution_name":          "Cascade Credit Union",
  "current_value":             "3234.11",
  "bank_stmt_avg_balance":     "2988.40",
  "manual_balance":            null,             // the household's own figure, when it wrote one
  "annual_income_from_assets": "12.40",
  "certification_form":        "HUD-50059",      // what the certification itself states
  "certification_cash_value":  "3234.11",        // for this asset: cash value and
  "certification_income":      "0.00",           // annual income (a printed 0 is sent)
```

```
   "interest_rate":             "0.004000",
   "percentage_of_ownership":   "100",
   "verification_status":       "received",

   "bank_statements": [
     { "statement_date": "2026-05-31", "balance": "3104.02",
       "source_label": "May statement" }
   ],
   "voa": { "voa_date": "2026-06-11", "reported_value": "3234.11",
            "source": "Cascade Credit Union",
            "month_1_balance": "2910.00", "month_2_balance": "3050.40",   // only when the VOA
            "month_3_balance": "2875.12", "month_4_balance": "3104.02",   // lists monthly balances
            "month_5_balance": "2990.00", "month_6_balance": "3234.11" }  // month 1 = oldest
 }
```

`month_1_balance` … `month_6_balance` are sent only when the verification of assets lists individual monthly balances instead of a six-month average (Chase and some others). Month 1 is the oldest; fewer than six months fills from month 1; more than six sends the most recent six. When they are sent, `bank_stmt_avg_balance` carries the mean of the months sent, so it agrees with the average Cartograph computes from them.

Never send `self_declared_balance`. It appears in the controller's permitted parameters but has **no backing column**. It is a seeded calculation-method code. Because records are built by mass assignment, including it raises `ActiveModel::UnknownAttributeError` and returns a 500. Use `manual_balance` for a declared balance and `self_declared_income` for declared asset income.

6b. checklist_matches

One entry per checklist row of the case the packet can answer, keyed on the row's own finding id from the case request (`checklist_rows[]`, or `requirements[].community_defaults.checklist_items[].case_finding_ids`). Rows the engine cannot map to a document type it knows are left out and stay untouched.

```
 { "finding_id": 88412, "found": true,  "pages": [2, 3],   "confidence": 0.85, "note_source": "scan",
   "note": "HUD 50059 present, pages 2-3. Not signed: the signature lines are blank." }
 { "finding_id": 88419, "found": true,  "pages": [38, 39], "confidence": 0.9,  "note_source": "scan",
   "note": "HUD 9887 present, pages 38-39. Signed by Yolanda Bribiesca on 2026-09-16." }
 { "finding_id": 88423, "found": true,  "pages": [40],     "confidence": 0.9,  "note_source": "scan",
   "note": "HUD 92006 present, page 40. Signed, date unclear." }
 { "finding_id": 88421, "found": false, "pages": [],       "confidence": 0.9,  "note_source": "scan",
   "note": "No HUD Race and Ethnic Data Form in the packet." }
```

`found: false` means the engine looked and the form is not in the packet. A row about a property of a form ("HUD 50059 Includes the Correct Income Limits") is not answered by the form being present; such rows are left out and the engine's findings cover them. A row asking for the previous form ("Previous HUD 50059") is left out: the case holds one certification and the prior year lives on its own record in Cartograph, so the packet is not expected to carry it (the engine answers such rows only when configured to expect a previous form). A form the engine knows to be incomplete says so in its note. A signed form's note gives the signature date when it was read; otherwise it says `date unclear` (something is written in the date slot that could not be read) or `undated` (the slot is blank) — a reading problem and a missing date are different things for the file. `confidence` is how sure the engine is of the match and the note together: 0.9 when the row and the document share a form number, 0.8 when the document's title is printed in the row's label, 0.6 when only words match; lowered when the classifier placed the page by nearest match or a signature could only be inferred. The note is written for the row's note field, with no marker in the text; `note_source: "scan"` on every entry is what tells it from a note staff typed. Cartograph records the source with the note, and on a later run clears a scan-sourced note from any row it sent that the reply omits, never one staff typed. The status stays Cartograph's to set.

7. expense_records

```
{ "member_ref": "m03", "expense_type": "dependent_care",
  "provider_name": "Bright Start Childcare", "annual_amount": "6240.00" }
```

Only these four keys are safe. `monthly_amount` and `verification_status` are permitted by the controller but have no columns and will 500 the request.

The engine does not currently extract expense data; this section is defined for completeness and will arrive empty until that capability exists (Section 15).

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8. Checklist determinations ride on findings

Cartograph dropped `cert_review_checklist_items` on 2026-08-18, replacing it with `CertChecklistTemplate` (the per-program item definitions, now carrying `item_code` and `cert_type_scope`) plus `CertReviewFinding` (the per-case result). There is no longer a separate table of per-review checklist rows, so this payload no longer carries a separate `checklist_items` array.

Instead, a finding that answers a checklist item carries that item's `item_code`. One finding per answered item; findings not tied to an item omit the key.

This requires one column that does not exist yet: `item_code` on `cert_review_findings`. `item_code` was added to the *template* table, not to findings, so a finding currently has no way to reference the item it answers. See the *Schema Change Request*, which lists this as the highest-priority addition.

Item codes are per-program and editable through the community funding program admin screen, so the engine cannot hardcode them. They must be supplied, either as an export or through the trigger payload. An `item_code` the engine does not recognise is left unanswered rather than guessed.

9. findings

Findings land in `cert_review_findings`, which already models subject, category, result, and assignment.

```
{
  "finding_key":  "INCOME_YTD_MISMATCH:m01:northgate",   // stable across re-imports
  "item_code":    "INC_FOUR_PLUS_PAYSTUBS",              // omit when not a checklist item
  "category":     "income",          // unit_rent | household_member | income |
                                     // asset | expense | file_review
  "subject_ref":  { "type": "income_record", "member_ref": "m01",
                    "source_name": "Northgate Logistics LLC" },
  "label":        "Paystub YTD does not match VOI YTD",
  "result":       "non_compliant",   // compliant | non_compliant | na
  "assignment":   "client",          // internal | client | procedural_issue
  "correction_required": "Obtain corrected VOI or a fourth paystub for this source",
  "confidence":   0.82,
  "resolution_type": "recalculation",   // presence_only | recalculation
  "pages":        [12, 13, 14],
  "position":     3
}
```

`subject_ref` is resolved by the importer to `subject_type` / `subject_id` against records created earlier in the same transaction. A finding with no subject (`type: "case"`) becomes a `file_review` finding with a null subject, matching Cartograph's existing convention.

`finding_key` is deterministic, derived from the finding type plus the member and source it concerns, so a re-import matches an existing finding rather than duplicating it. **Findings whose key and label are unchanged retain their existing result and any reviewer resolution.** Only findings whose determination changed are reset.

`confidence`, `resolution_type`, and `pages` require the new columns listed in the *Schema Change Request*. They drive auto-approve gating and reviewer guidance respectively.

---

10. Controlled vocabularies

Two are enforced by model validation; the rest are convention but should still be normalized so filters behave.

10.1 income_type (validated)

Engine value	Cartograph value
Non-Federal Wage	`non_federal_wages`
Federal Wage	`employment` (closest available; see note below)
Social Security	`social_security`
Supplemental Security Income	`ssi`
Social Security Disability	`social_security`
Pension	`pension`
Temporary Assistance	`public_assistance`
Child Support	`child_support`
Self-Employment	`self_employment`
Zero Income	`other` (+ zero_income child)
Other Income	`other`

No federal-specific value exists, so federal wages map to the generic `employment`. Confirm this is the intended target before go-live.

Cartograph accepts nine further values the engine does not currently emit: `alimony`, `wages_and_salaries`, `income_in_cash`, `unemployment_benefits`, `armed_forces_pay`, `veterans_benefits`, `investments`, `real_estate_income`, `student_financial_aid`.

10.2 asset_type (validated)

Cartograph allows nine values: `checking`, `savings`, `money_market`, `cd`, `stocks`, `bonds`, `real_estate`, `retirement`, `other`.

Engine value	Cartograph value
Checking	`checking`
Savings	`savings`
CD	`cd`
Investment	`stocks`
Retirement	`retirement`
Real Estate	`real_estate`
Life Insurance, Cryptocurrency, Prepaid Card, Peer-to-Peer, ABLE Account, Cash, Annuity, Direct Express	`other`

Eight engine values collapse to `other`. The original is preserved in `notes` and reported in the result callback's `warnings`.

10.3 Remaining vocabularies

Field	Accepted values
pay / income frequency	`weekly` `biweekly` `semimonthly` `monthly` `annually`
`verification_status`	`pending` `requested` `received` `waived`
`expense_type`	`dependent_care` `disability_assistance` `medical` `other`
finding `result`	`compliant` `non_compliant` `na` `resolved`
finding `assignment`	`internal` `client` `procedural_issue`
checklist `result`	`pending` `pass` `fail` `na`
issue `severity`	`blocking` `warning` `info`
`relationship`	Head of Household · Spouse · Co-Head · Minor Child · Foster Child/Adult · Live-in Aide · Other Adult Member

---

11. Import order

The whole import runs in one transaction. Order matters because children carry foreign keys to records created earlier in the same request.

1. Verify signature and timestamp, before parsing the body.
2. Resolve the job, then find-or-create its cert review. A missing job is a `409`: the packet is valid, the target is not ready.
3. Open the transaction and clear prior **extraction-owned** child records (see below).
4. Update `cert_review` scalars, using only keys present in the payload. Leave `result`, `signed_off_at`, and `notes` untouched.
5. Create household members, building the `ref` → ID map.
6. Create income records, then their paystub / VOI / zero-income children. An unresolvable `member_ref` is a hard failure, not a null.
7. Create asset records, then bank statements and VOA. Then expense records.
8. Upsert findings by `finding_key`, carrying `item_code` where the finding answers a checklist item.
9. Commit, then run `CertIncomeCalculationService`. Running after commit ensures it reads settled data.
10. Post the result callback.

"Extraction-owned" requires a marker

Full-replace in step 3 is only safe if the delete can distinguish records this endpoint created from records an analyst entered by hand. Cartograph has no such marker today. Blind-deleting every `cert_income_record` on re-import would destroy reviewer work.

The *Schema Change Request* asks for a `source` column defaulting to `"manual"`, so the delete becomes `where(source: "extraction")`. **Until that exists, treat the endpoint as create-only against an empty cert review and reject a second import with `409`.**

---

12. Income calculations stay out of the payload

Cartograph computes its own into `cert_income_calculations`, and its method list (`self_declared`, `paystub_avg`, `paystub_max`, `paystub_ytd`, `voi`, `voi_ytd`) does not align with the engine's four internal methods. The engine sends raw paystubs and VOIs and lets Cartograph calculate.

The engine still computes income internally, since that is how it produces findings. Where the two diverge materially, the engine raises the divergence as a finding rather than writing a competing calculation row.

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13. SSN handling

This contract carries `ssn_last4` and no full SSN field, deliberately. Cartograph's `ssn` column is plain text with no Active Record encryption declared in the model layer. Sending full digits would move them from a masked, controlled store into an unencrypted one.

If full write-back is needed later, the precondition is `encrypts :ssn` on `CertHouseholdMember` plus a backfill; the field would be added at v1.2, not before.

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14. Corrections from earlier versions

Rows 1 to 6 were corrected in v1.1 against the repository; rows 7 to 9 in v1.2 against the deployed endpoint.

#	v1.0 assumption	Correct behaviour
1	Findings go to `cert_review_issues`	`cert_review_findings`, which models polymorphic subject, category, result enum, and assignment
2	Checklist items are created/replaced	Auto-seeded by `after_create :seed_checklist`; upsert by `item_code`, unique index on `(cert_review_id, item_code)`
3	Severity values `warning / error / info`	`blocking / warning / info`. `error` fails validation
4	`self_declared_balance` is a column	It is a seeded calculation-method code, not a column. `self_declared_income` and `manual_balance` are the real columns
5	`percentage_of_ownership` and real-estate values have no column	All three exist on `cert_asset_records`
6	`cert_type` mapping complete	Only three values exist; AR-SC unmapped
7	Checklist determinations upsert into `cert_review_checklist_items`	That table was dropped 2026-08-18. Determinations ride on findings via `item_code`, which findings do not yet have
8	Signature is `v1,<base64>`, headers `X-Idp-*`	Deployed endpoint expects a **hex** digest with no prefix, on `X-Timestamp` / `X-Signature`
9	Routing on `target.job_id`	Controller resolves `case_ref` against `Job#ref_number`

---

15. Open decisions

1. `item_code` **list.** Codes are per-program and editable, so the engine cannot hardcode them. Supply them as an export or in the trigger payload.
2. **AR-SC.** `cert_type` has three values and the engine handles four. Add a value, or map AR-SC onto `annual` with the distinction carried elsewhere?
3. **Packet definition.** Which attachment set constitutes the packet: `JobDocument#document` filtered by `document_class`/`source`, or `CertReview#documents`?
4. **Expense extraction.** Expenses affect adjusted income on HUD certifications. Should the engine extract them, or do they remain staff-entered?
5. **Income authority.** Confirmed as Cartograph computing, with engine divergence surfaced as a finding. Recorded here so it is explicit rather than assumed.